

FLIPPED CLASSROOM — SESSION 4

Clustering Analysis & Data Visualization

Data Analysis & Pattern Recognition — Exploratory Data Analysis Module

This is a flipped-classroom session: you prepare the content before class, and class time is used for discussion, live problem-solving, and presenting your work — not for a first introduction to the topic. Make sure you complete everything below before the session.

Learning objectives

By the end of this session, you should be able to:

- Apply k-means, hierarchical (agglomerative) clustering, and Gaussian Mixture Models to a real dataset.
- Prepare data for clustering: select numeric variables and handle missing values (reusing pandas techniques from S2).
- Visualize and interpret clustering results with seaborn (scatterplot, pairplot).
- Evaluate clustering quality both against a known ground truth (confusion matrix, Adjusted Rand Score) and without one (BIC).
- Apply basic Python skills (functions, loops) by implementing a Euclidean distance function and a dictionary-based counter from scratch.

Before class — checklist

Work through this on your own, before the session. Budget enough time — this is not a 20-minute task.

- ☐ Review the theory notebook S3. Clustering analysis PRML (k-means, hierarchical/agglomerative clustering, GMM, model comparison, choosing the number of clusters with BIC).
- ☐ If needed, briefly review S1 (functions, loops) and S2 (pandas: filtering, missing values, groupby; seaborn: scatterplot, pairplot).
- ☐ Answer the 3 self-check questions below, in your own words (no need to submit them).
- ☐ Complete the 3 exercises in S4_FlippedClassroom_Exercises_Student.ipynb.
- ☐ Submit your notebook on ATENEA and share your GitHub link, before your group's deadline (see Submission below).

Self-check questions

Try to answer these, even intuitively, before moving on to the exercises:

- What is the main difference between k-means and hierarchical clustering in terms of how the number of clusters is decided?
- Why can k-means fail on non-spherical or very unevenly-sized clusters?
- What is the BIC criterion used for in a Gaussian Mixture Model (GMM)?

If you can't answer these after reviewing the theory, don't worry — just note it down. We'll start class with exactly these questions.

The 3 exercises

No solutions are provided on purpose. The goal is to try, get stuck at some point, and bring your specific questions to class.

Exercise 1 — K-means on the Penguins dataset

(Integrates S2 + S3)

- Select the numeric columns and handle missing values (review from S2).
- Apply k-means with 3 clusters.
- Visualize the result with a scatterplot colored by cluster.
- Compare the result against the real species using a confusion matrix and the Adjusted Rand Score (new, from S3).

Exercise 2 — Hierarchical clustering and a manual distance function

(Integrates S1 + S3)

- Implement, in plain Python (no numpy), a function that computes the Euclidean distance between two points using a for loop (review from S1).
- Apply hierarchical clustering and plot the dendrogram.
- Apply k-means on the same data.
- Compare both results, including a function that counts observations per cluster using a dictionary (review from S1).

Exercise 3 — Choosing the number of clusters with GMM and BIC

(Integrates S2 + S3)

- Select the relevant numeric variables from the diamonds dataset (used in S2).
- Fit a GMM for different numbers of components and plot the BIC curve.
- Identify the optimal number of clusters according to BIC.
- Visualize the final result with a pairplot colored by cluster (review from S2).

On using AI tools

You may use AI tools to help complete the exercises. If you do, you must:

1. Justify your solution in detail (explain your reasoning, not just paste code).
2. Be able to explain the results in your own words when asked in class.

Submission

Deadlines are per lab group — check which one applies to you:

Group	Deadline	Time
T11, T12, T22	Tuesday, 13th	12:00
T21	Thursday, 15th	12:00

- Submit your completed notebook on ATENEA before your group's deadline above, as a .ipynb file (not a PDF or a zipped folder).
- Also include a link to your GitHub repository, so your results can be easily reviewed.

Materials

File	What it's for
S3. Clustering analysis PRML.ipynb	Theory notebook (k-means, hierarchical clustering, GMM, evaluation). Review it before starting the exercises. Included in the session folder.

File	What it's for
S4_FlippedClassroom_Exercises_Student.ipynb	The exercises notebook you need to complete — this is your pre-class homework.
S1. Intro Python.ipynb / S1 Exercises	Refresher on functions and loops, needed for Exercise 2.
S2 Data Visualization.ipynb / S2 Exercises	Refresher on pandas, missing values, and seaborn plots, needed for Exercises 1 and 3.

Which software can I use?

You can use whatever software you're comfortable with to run the notebook. We recommend having Jupyter Notebook installed locally (or any way of running notebooks locally that works for you), but Google Colab is also perfectly fine.

What happens in class

Class time is for discussion and troubleshooting, not for a first introduction to the topic — come having already attempted all 3 exercises.

- Three students will be chosen at random, one per exercise, to present their solution to Exercise 1, Exercise 2, or Exercise 3 and walk the class through their reasoning.
- After each presentation, the group will discuss the solution together — alternative approaches, common mistakes, and open questions.
- The instructor will then present the reference solution for each exercise, highlighting the key ideas and connecting them back to S1, S2, and S3.

Because anyone can be selected to present, make sure you can explain your own solution — not just that it runs, but why it works.